

1. Problem Statement

The Employee Burnout Prediction System is a software application developed to predict and monitor employee burnout risk using employee-related information and machine learning models. The system allows HR Managers to manage employee data, generate burnout predictions, view prediction history and trends, and identify high-risk employees. Administrators manage users and datasets, while Machine Learning Engineers train, retrain, and update the prediction models.

2. Identified Actors**2.1 HR Manager**

The HR Manager uses the system to manage employee information and monitor employee burnout.

Responsibilities:

- Login
- Manage Employee Data
- Predict Burnout Risk
- View Prediction Result
- View Prediction History
- View Burnout Trend
- View High-Risk Alert
- Generate Reports

2.2 Administrator

The Administrator manages system users, employee data, datasets, and machine learning models.

Responsibilities:

- Login
- Manage Employee Data
- Manage Users
- Manage Dataset
- Update Model

2.3 Machine Learning Engineer

The Machine Learning Engineer manages the machine learning model and training activities.

Responsibilities:

- Login
 - Manage Dataset
 - Train Model
 - Retrain Model
 - Update Model
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3. List of Use Cases

The major use cases identified from the system are:

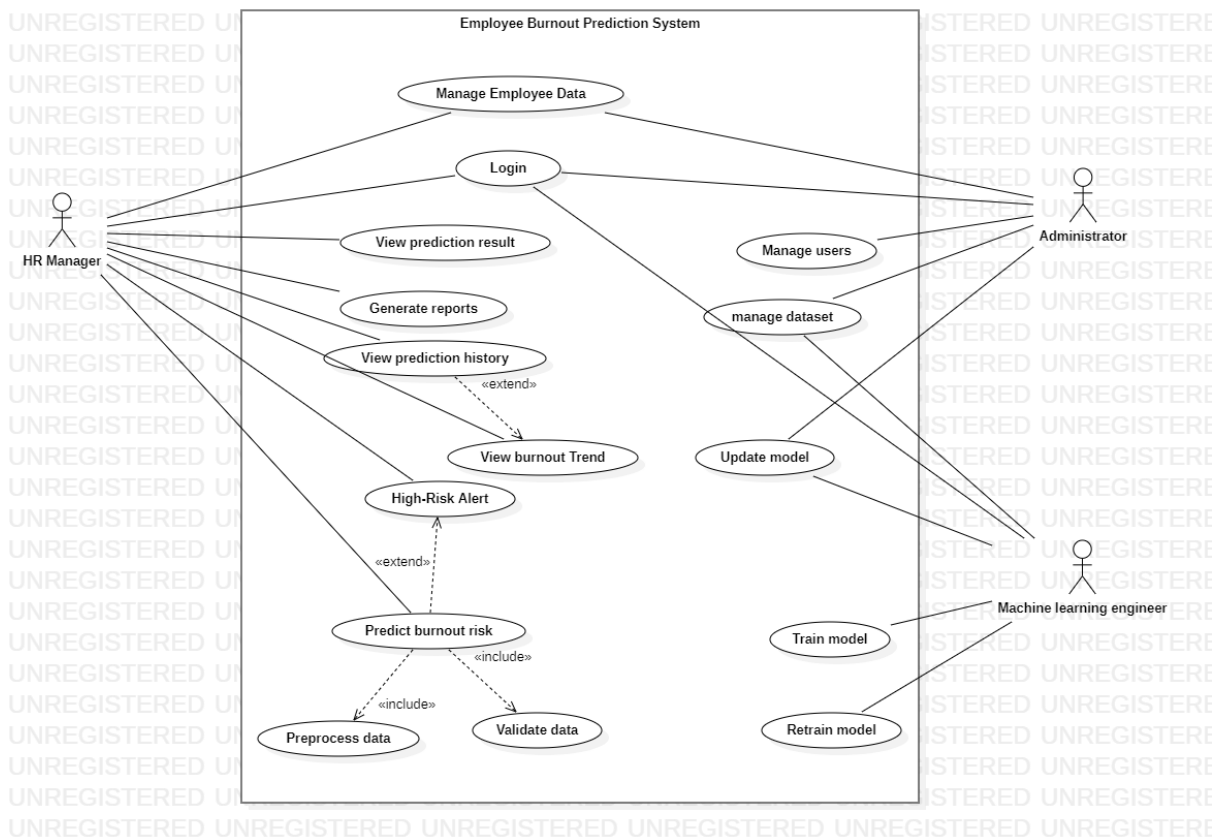
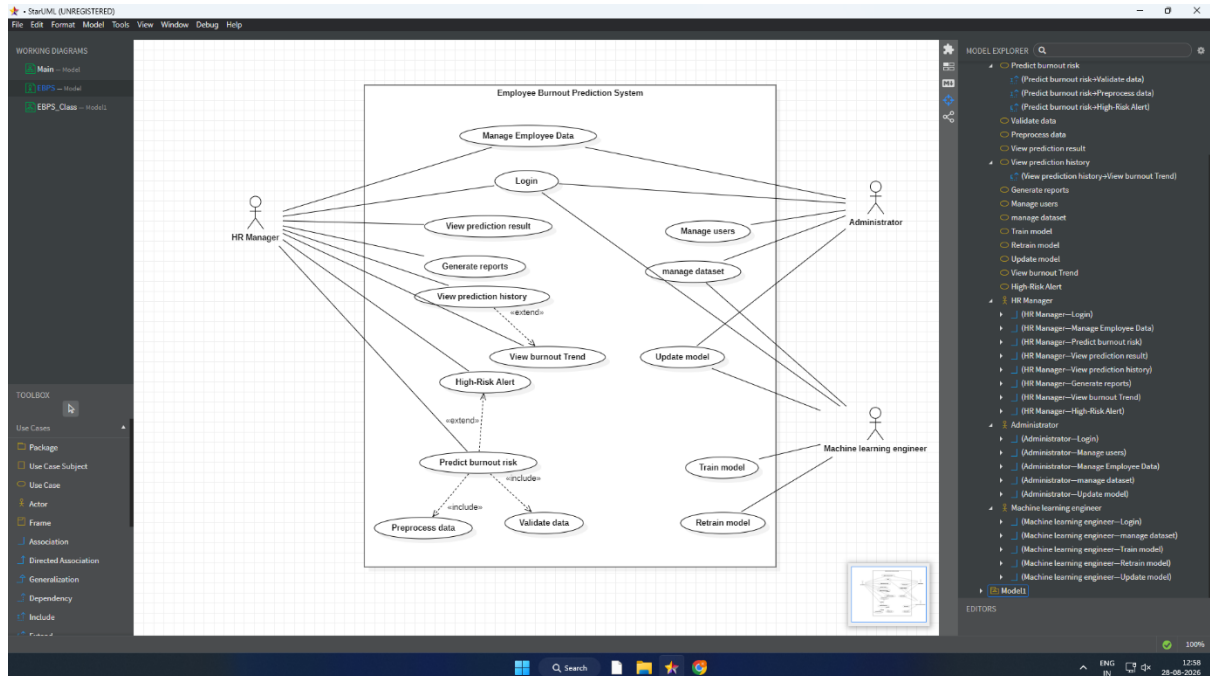
1. Login
2. Manage Employee Data
3. Predict Burnout Risk
4. Preprocess Data
5. Validate Data
6. View Prediction Result
7. View Prediction History
8. View Burnout Trend
9. High-Risk Alert
10. Generate Reports
11. Manage Users
12. Manage Dataset
13. Train Model
14. Retrain Model
15. Update Model

Use Case Relationships

- Predict Burnout Risk <<include>> Preprocess Data
- Predict Burnout Risk <<include>> Validate Data
- View Burnout Trend <<extend>> View Prediction History
- High-Risk Alert <<extend>> Predict Burnout Risk

4. Use Case Diagram

Figure 1: Use Case Diagram of Employee Burnout Prediction System



Brief Explanation

The Use Case Diagram represents the interaction between the system and its three main actors: HR Manager, Administrator, and Machine Learning Engineer. HR Managers perform employee management and burnout monitoring, Administrators manage users and system data, and Machine Learning Engineers manage datasets and machine learning models. The diagram also shows include and extend relationships for prediction, preprocessing, validation, trend analysis, and high-risk alerts.

5. Identified Classes

The following classes were identified for the Employee Burnout Prediction System:

1. Employee
 2. User
 3. Prediction
 4. Model
 5. PredictionFeature
 6. BurnoutProfile
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6. Class Attributes and Operations

6.1 Employee

Attributes:

- employeeID : int (PK)
- dateofJoining : Date
- gender : String
- wfhSetup : String
- designation : int
- resourceAllocation : float
- mentalFatigueScore : float

Operations:

- addEmployee()
 - updateEmployee()
 - getEmployee()
-

6.2 Model

Attributes:

- modelID : int (PK)
- modelName : String
- version : String
- trainingDate : Date
- accuracy : float
- status : String

Operations:

- trainModel()
 - evaluateModel()
 - updateModel()
-

6.3 User

Attributes:

- userID : int (PK)
- name : String
- email : String
- password : String
- role : String

Operations:

- login()
 - logout()
-

6.4 Prediction

Attributes:

- predictionID : int (PK)
- employeeID : int (FK)
- modelID : int (FK)
- userID : int (FK)
- predictionDate : Date

- riskScore : float
- riskLevel : String

Operations:

- predictionBurnout()
 - getResult()
-

6.5 PredictionFeature

Attributes:

- featureID : int (PK)
- predictionID : int (FK)
- mentalFatigue : float
- resourceAllocation : float
- designation : int
- month : int

Operation:

- prepareFeatures()
-

6.6 BurnoutProfile

Attributes:

- profileID : int (PK)
- currentRiskLevel : String
- lastAssessmentDate : Date
- monitoringStatus : String
- employeeID : int (FK)

Operations:

- updateProfile()
 - getRiskLevel()
-

7. Primary Keys and Foreign Keys

Primary Keys

Class	Primary Key
Employee	employeeID
User	userID
Prediction	predictionID
Model	modelID
PredictionFeature	featureID
BurnoutProfile	profileID

Foreign Keys

Class	Foreign Key
Prediction	employeeID → Employee
Prediction	modelID → Model
Prediction	userID → User
PredictionFeature	predictionID → Prediction
BurnoutProfile	employeeID → Employee

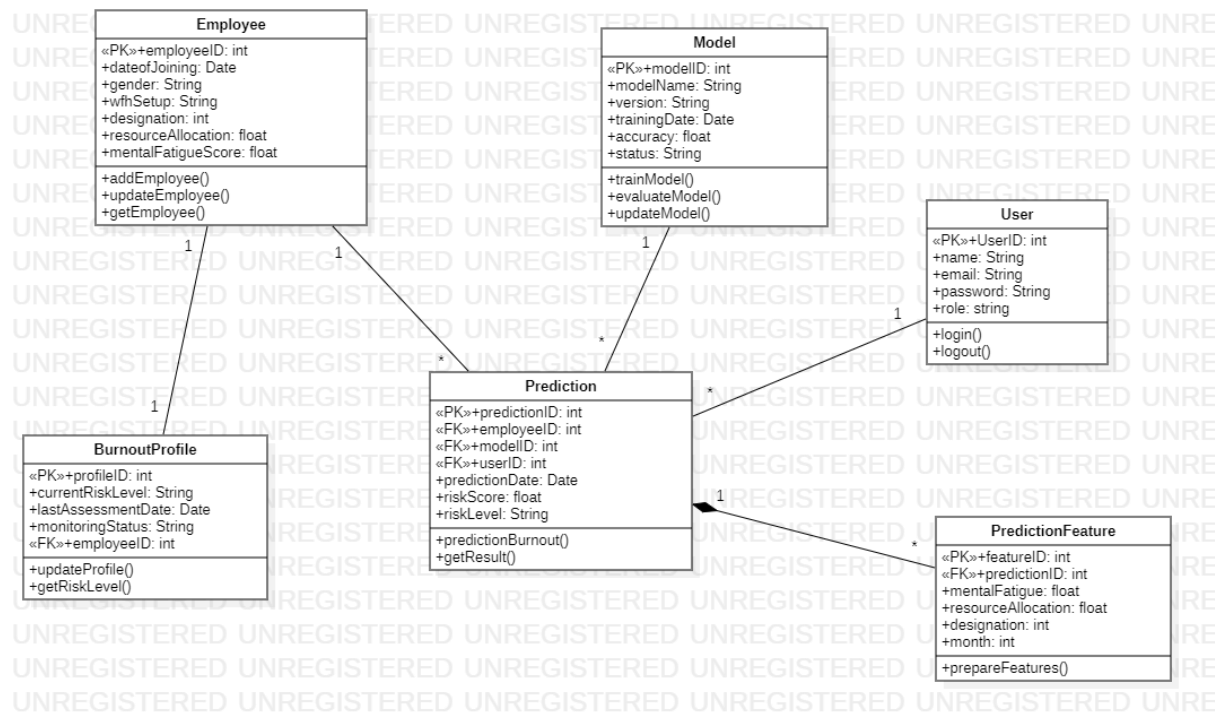
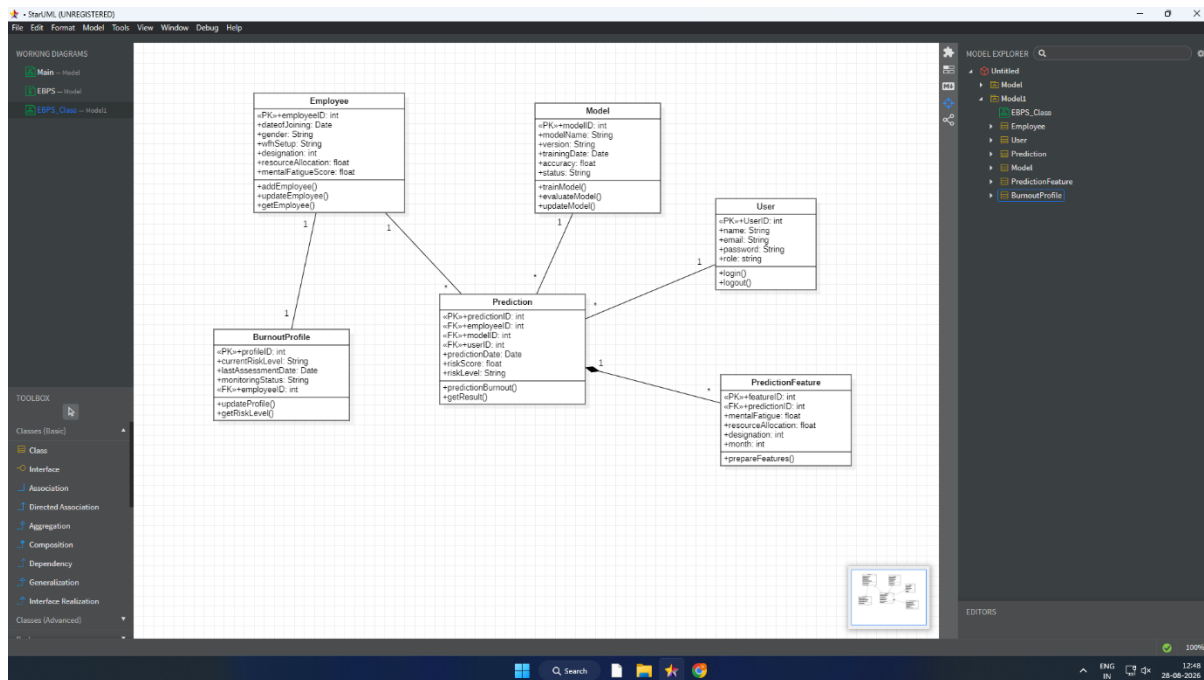
8. Relationships and Cardinality

Relationship	Cardinality Description	
Employee — Prediction	1 : M	One employee can have multiple predictions.
User — Prediction	1 : M	One user can generate multiple predictions.
Model — Prediction	1 : M	One model can generate multiple predictions.
Prediction — PredictionFeature	1 : M	One prediction can contain multiple features.
Employee — BurnoutProfile	1 : 1	One employee has one burnout profile.

The Prediction–PredictionFeature relationship is represented using composition in the Class Diagram because prediction features belong to a particular prediction.

9. Class Diagram

Figure 2: Class Diagram of Employee Burnout Prediction System



Brief Explanation

The Class Diagram represents the static structure of the Employee Burnout Prediction System. The Employee and User classes store employee and user information, while Prediction stores generated burnout prediction results. The Model class represents the machine learning model, PredictionFeature stores the features used for prediction, and BurnoutProfile maintains the

employee's burnout monitoring information. Primary keys, foreign keys, methods, and class relationships are represented to provide a structured object-oriented design.

10. Brief Explanation of Design Decisions

The system is divided into separate classes according to their responsibilities. Employee manages employee information, User manages system users, and Prediction stores the results generated by the machine learning model. Model manages model-related information, while PredictionFeature stores the input features used during prediction. BurnoutProfile maintains the current burnout status of an employee. The use case model and class model together provide a clear representation of the system's functional and structural requirements.